



Gabriela Bieda

Contact

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Nationality

Poland

Strengths

- Analytical thinking
- Problem solving
- Research oriented mindset
- Ability to work in interdisciplinary teams
- Scientific communication
- Quick learning
- Good organization and responsibility

Research Interests

Computational biology, bioinformatics, microbiome and metagenomic data analysis, machine learning for biomedical applications, generative models, biological sequence data, molecular modelling and precision medicine

Portfolio

<https://gabii3355.github.io/>

Profile

MSc student in Biosciences with a specialization in Bioinformatics at Wrocław University of Science and Technology, with a background in computational biology, molecular modelling, Python based data analysis and machine learning. My current MSc thesis focuses on modelling the spontaneous deamination of cytosine in DNA and the impact of nucleotide sequence context, which developed my interest in sequence based biological data, computational modelling and quantitative interpretation of biological mechanisms. I am particularly interested in applying AI and data science methods to biomedical data, including microbiome and metagenomic datasets.

Achievements and certificates:

- ◆ Participation in the "Frontiers in Medicinal Chemistry" conference, Munich 2024
- ◆ Rector's scholarship awarded in the academic years 2022-2026 for outstanding academic achievements, as well as the Santander scholarship for students of the Krakow University of Technology
- ◆ 2nd place in the Departmental Scientific Circle Session
- ◆ Participation in the course "The complete Python Bootcamp From Zero to Hero"
- ◆ Participation in the course "Bioinformatics; Learn Docking & Molecular Dynamics Simulation" and "Basics of JavaScript – Browser Programming Language"
- ◆ Participation in "Unite! Research School" in Politecnico di Torino and Eastern European Bioinformatics and Computational Genomics Workshop EEBG 2026

Skills:

- ◆ Programming and Data Analysis
 - ✓ Python, R, NumPy, pandas, scikit learn, TensorFlow, statistical analysis, data preprocessing, model evaluation, parsing and validation of computational outputs
- ◆ Computational Chemistry and Molecular Modelling
 - ✓ Molecular dynamics, docking, QM calculations, force field based simulations, molecular geometries, interaction analysis, RMSD analysis, HPC and Linux workflows
- ◆ Computational Chemistry Tools
 - ✓ GROMACS, LAMMPS, ORCA, AutoDock Vina, NAMD, VMD, PyMOL, Gaussian, Maestro, Chimera, Avogadro, Molden
- ◆ AI and Machine Learning
 - ✓ Machine learning for biomedical and chemical data, generative models, foundation models, model evaluation, data preprocessing, reproducible Python based workflows

Research Experience

MSc Thesis: Spontaneous deamination of cytosine in DNA. Modelling the impact of nucleotide sequence

- Investigating how local nucleotide sequence context may influence cytosine deamination in DNA
- Working with computational models of DNA systems and comparing sequence variants
- Applying quantitative analysis to interpret interaction patterns and differences between molecular states
- Developing experience in biological interpretation of computational outputs and sequence based reasoning
- Strengthening skills in data organization, reproducibility and scientific reporting

Bachelor Thesis "Production of selected floral absolutes via room-temperature extraction"

- Performed quantitative analysis of GC FID datasets, including calibration modelling and result validation
- Processed and interpreted experimental data to identify trends and concentration profiles
- Documented structured analytical results to ensure reproducibility and consistency

Participation in the scientific project Future Lab "TryPto Holder - A new method for treating pancreatic cancer using compounds that inhibit serotonin biosynthesis by blocking the TPH1 enzyme" and "New ligands of the 5-HT₆ receptor from the group of sulfonamide derivatives of cyclic aryloguanidines in the treatment of glioblastomas" + Sulfonamide derivatives of 1,4-dihydroquinazolin-2-amine as a new scaffold for selective 5-HT₆R ligand design

- Prepared and standardized molecular datasets for computational modelling workflows
- Performed comparative analysis of candidate compounds using docking simulations and quantitative scoring metrics
- Evaluated interaction profiles and stability indicators to support data driven selection of candidates
- Analyzed simulation outputs, including RMSD trends and interaction persistence
- Worked with HPC based computational outputs and organized results into structured reports and visual summaries

Work experience

01/11/2024- 30/01/2025

Internship at Selvita S.A - "Selvita Chemistry Academy", Cracow

Position: Laboratory employee

Tasks:

- ◆ Performing basic synthesis reactions in the pharmaceutical industry, including: creating amides, Witting reaction, Buchwald reaction, and Pictet-Spengler reaction
- ◆ Operating equipment and analyzing H-NMR and LC-MS results
- ◆ Searching for literature related to the above chemical reactions
- ◆ Selecting reaction parameters (temperature, time, reagents)

01/07/2024- 09/08/2024

Internship at Euphora Paese Sp. z o. o., Jawornik

Position: Employee in the Quality Control, Production and Laboratory departments

Tasks:

- ◆ Production of cosmetics at the laboratory level
- ◆ Analysis of raw materials, manufactured masses and verification of the quality of the final product
- ◆ Product labeling
- ◆ Packing of goods

Education

Studies

01/03/2025-10/07/2026)

Wrocław University of Science and Technology

MSc studies in Bioscience (in English), Specialization: Bioinformatics

Studies

1/10/2021-03/02/2025

Cracow University of Technology

BSc studies in Chemical Technology, Specialization: Light Organic Technology

Languages

- ◆ Polish - native
- ◆ English - C1
- ◆ Spanish - B1
- ◆ German - A1

Clause

I agree to the processing of personal data provided in this document for realising the recruitment process pursuant to the Personal Data Protection Act of 10 May 2018 (Journal of Laws 2018, item 1000) and in agreement with Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data, and repealing Directive 95/46/EC (General Data Protection Regulation).